DO NOW — RECALL

# Lesson 107 · Paging, Segmentation and Virtual Memory

Name: _____ Class: _____ Date: _____

**1. Identify two services provided by an OS.**

**2. What is the time complexity of sequential search in the worst case?**

**3. After pushing A then B onto a stack, which is popped first?**

**4. Why must a binary search operate on ordered data?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Any two: process management, memory management, files, devices, security or interface.

### 2. Model answer

O(n).

### 3. Model answer

B, because a stack is LIFO.

### 4. Model answer

It discards half the search space using an ordering comparison.

### By the end, you can…

• Virtual memory is split into fixed-size pages; RAM into frames.

• Logical variable-size segments represent code, data or stack regions.

• The OS maps addresses and can move inactive pages to secondary storage.

### Vocabulary — know these terms

page — Fixed-size virtual-memory block.

frame — Fixed-size physical-RAM allocation unit.

segmentation — Memory organised into variable-size logical segments.

page fault — Access to a page not currently resident in RAM.

NEW LEARNING

# Read, annotate and explain

### Paging

Virtual memory is split into fixed-size pages; RAM into frames.

### Segmentation

Logical variable-size segments represent code, data or stack regions.

### Virtual memory

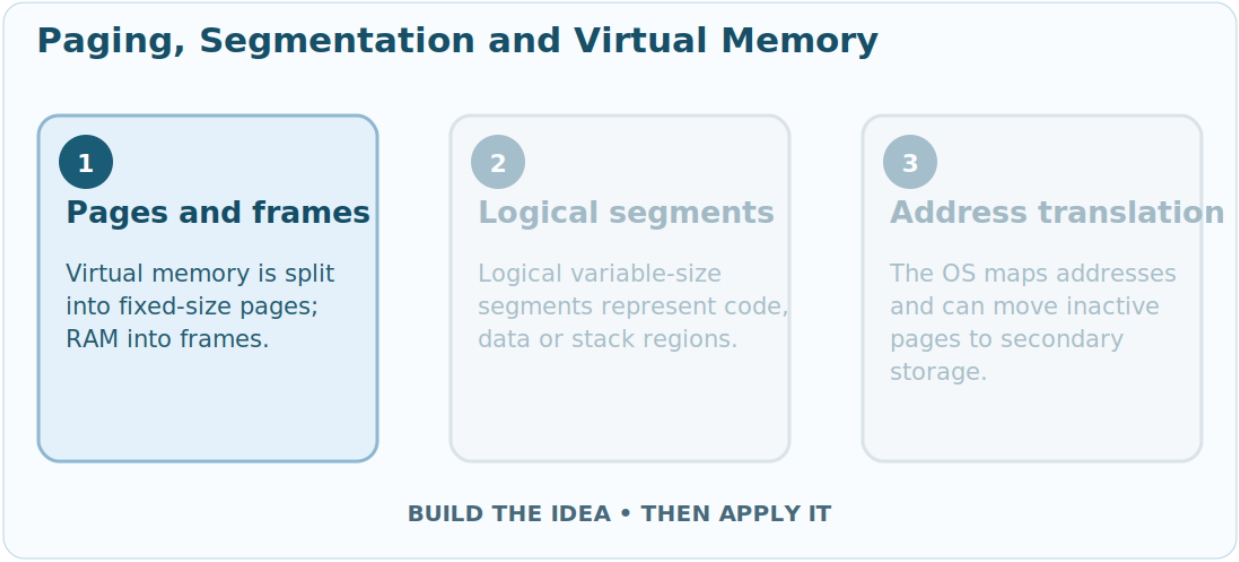
The OS maps addresses and can move inactive pages to secondary storage.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

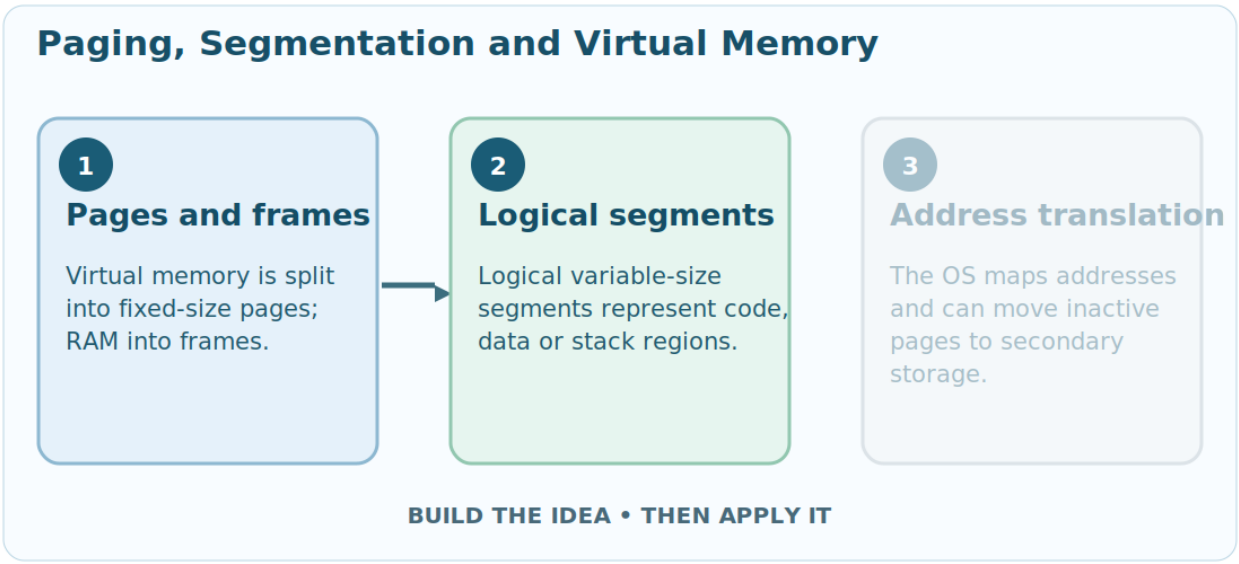
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Paging, Segmentation and Virtual Memory: Pages and frames then Logical segments then Address translation

### New Learning

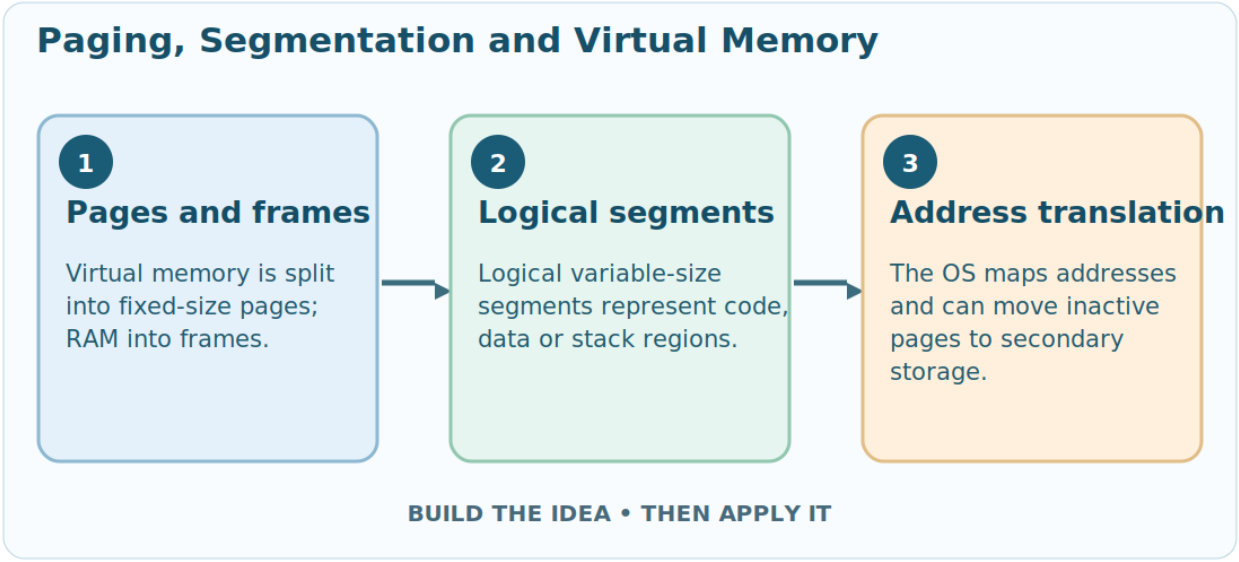



Paging, Segmentation and Virtual Memory: Pages and frames then Logical segments then Address translation

# New Learning · original lesson diagrams

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**New Learning**




Paging, Segmentation and Virtual Memory: Pages and frames then Logical segments then Address translation

I DO — WORKED EXAMPLE

# Follow the worked method

**A 2,500-byte virtual address uses 1,024-byte pages. Find the page and offset, then place it in frame 7.**

## Model answer

Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7.

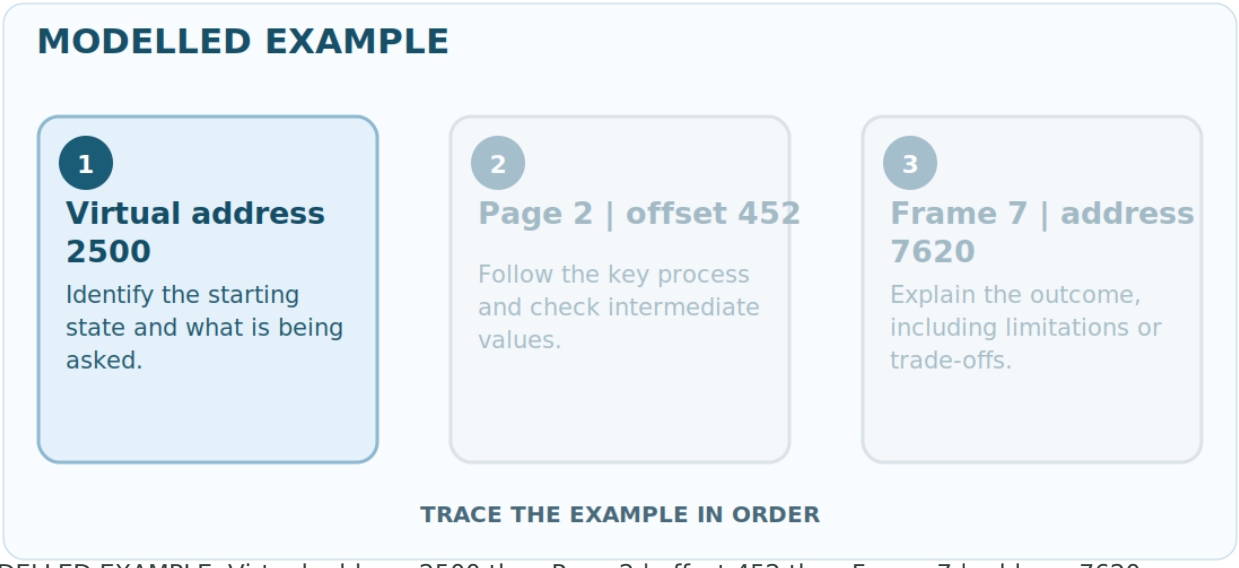
Check: link each claim to the scenario and use precise OCR terminology.

# I Do · original lesson diagrams

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Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7.




MODELLED EXAMPLE: Virtual address 2500 then Page 2 | offset 452 then Frame 7 | address 7620

# I Do · original lesson diagrams

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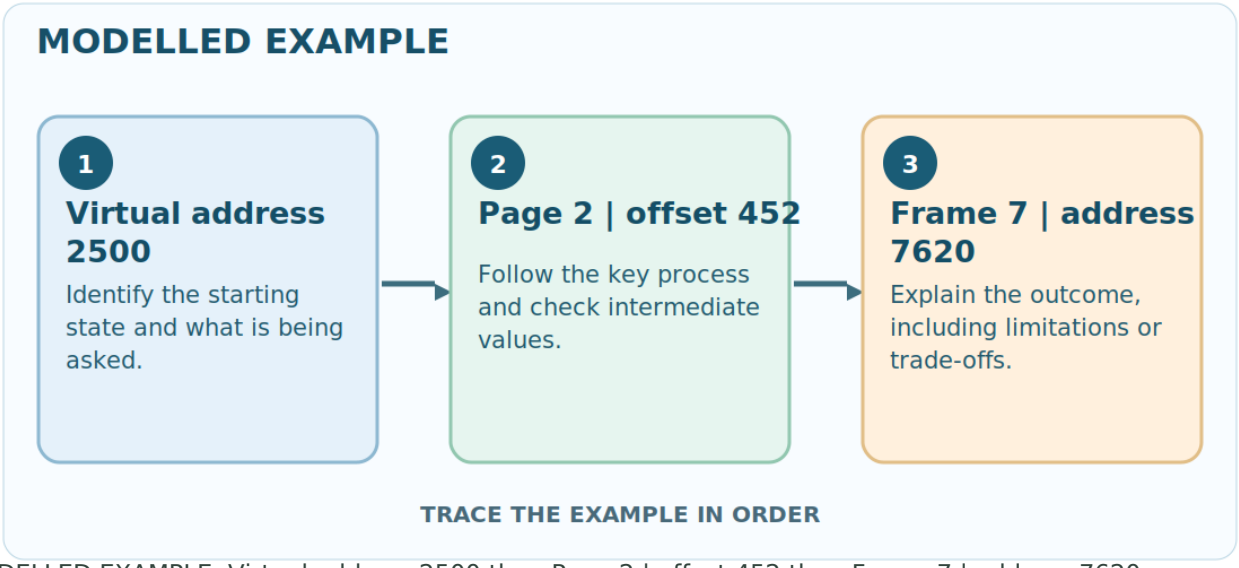

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WE DO — GUIDED PRACTICE

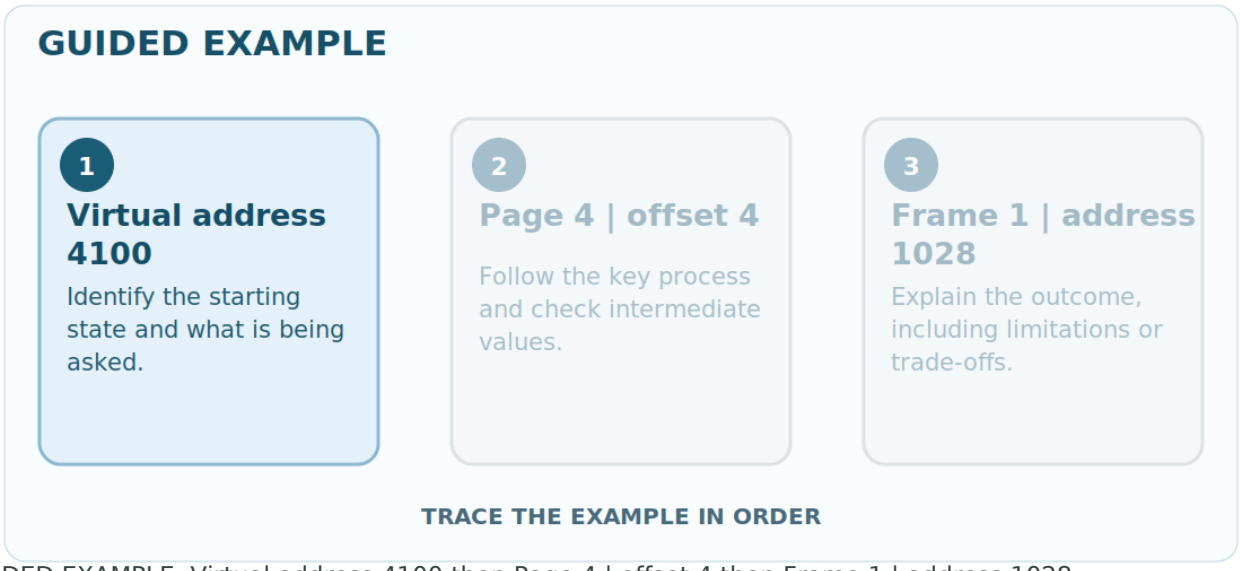
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A 4,100-byte virtual address uses 1,024-byte pages; page 4 maps to frame 1. Find the physical address.**

Identify the inputs, relevant facts and exact output or judgement required.




GUIDED EXAMPLE: Virtual address 4100 then Page 4 | offset 4 then Frame 1 | address 1028

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Differentiate paging and segmentation, with one reason for using each. [6]

### Question 2

With 4,096-byte pages, calculate page number and offset of virtual address 8,193. [4]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why heavy paging can cause thrashing and slow down a computer. [8]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Virtual memory increases the speed of every memory access.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

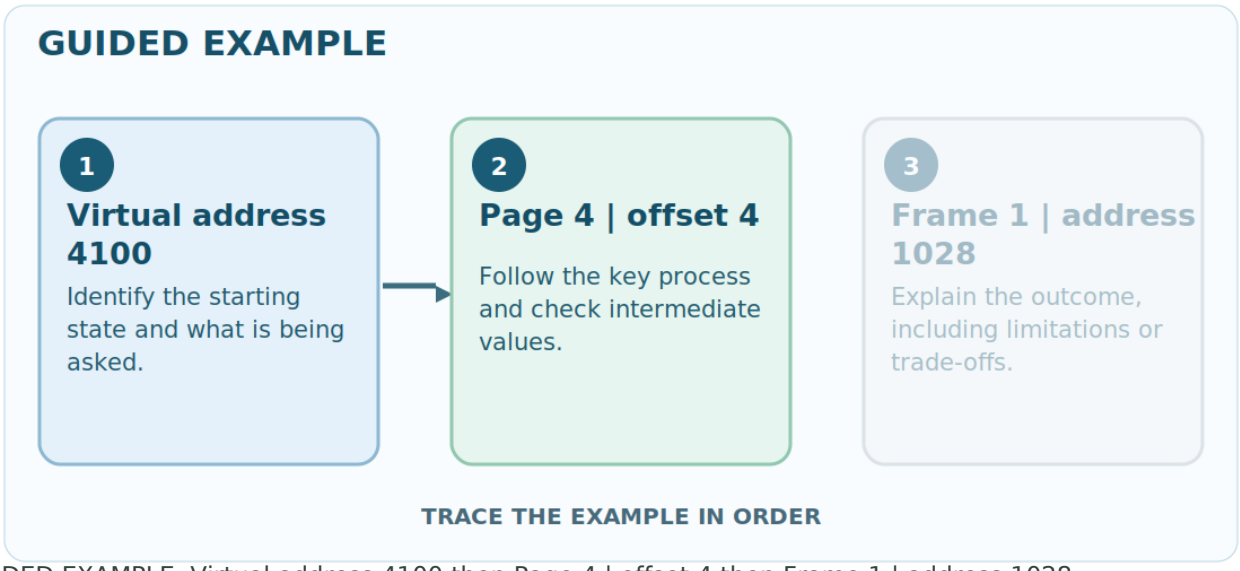
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

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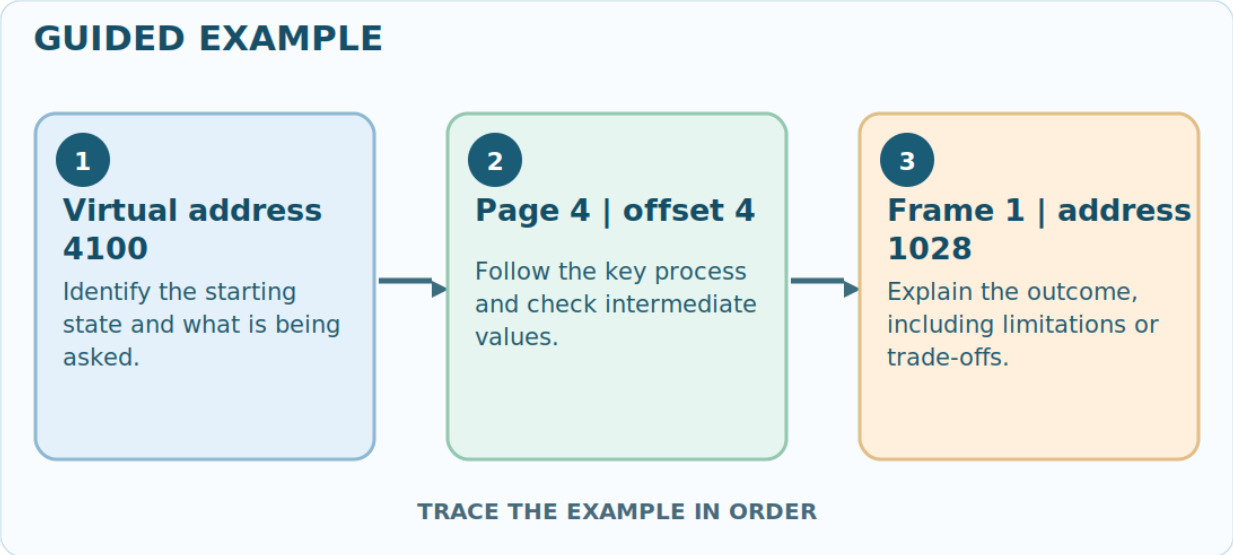

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# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

4,100 ÷ 1,024 gives page 4 and offset 4. Frame 1 starts at 1,024 so physical address = 1,028.

IMPROVE: Explain one correction to your own reasoning.